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ELECTROSTATIC DISCHARGE INSOLE

Abstract

The present invention discloses an advanced insole for electrostatic discharge footwear. The insole features a base foam layer and a fabric top cover with conductive threads woven into the cover material, providing electrical flow paths from a wearer's foot to the base foam layer. An electrostatic discharge chemical additive is incorporated into the base foam layer, creating polarity channels for an electrical pathway within the foam layer. The insole is designed in a 3D supportive shape for increased foot comfort and interface, including a heel cup and arch support. The edges adjacent the toe region are beveled to accommodate various footwear, and the bottom surface of the base foam layer includes molded-in structures that provide varying degrees of cushioning to different regions of the foot. This novel insole design enhances comfort while ensuring efficient electrostatic discharge, proving advantageous in various footwear applications.

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Background/Summary

PRIORITY CLAIM [0001] This application claims priority to U.S. Provisional Application Ser. No. 63/557,765 filed Feb. 26, 2024, entitled INSOLE WITH ELECTROSTATIC DISCHARGE, which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION
[0002] This application relates to footwear insoles and, more specifically, to insoles having electrostatic discharge properties.
BACKGROUND OF THE INVENTION
[0003] The present invention pertains to the field of footwear that provides low resistance to electrostatic discharge, such as might be needed in some work environments. More specifically it is related to the design of insoles capable of electrostatic discharge. Electrostatic discharge footwear requires an insole or sock liner capable of transmitting a build-up of static electricity to the ground. This necessitates a path through the insole to a last board, midsole, and outsole that are also capable of electrostatic discharge.
[0004] A major challenge in this field is the construction of an insole that provides an efficient electrostatic discharge pathway while maintaining comfort for the wearer. Previous attempts have primarily focused on stitching conductive threads through the insole. These threads must extend through all layers of the insole, including the foam layer and the top cover. However, this approach creates small ridges and recesses that interface with the foot, leading to discomfort as the wearer presses full weight onto the insole top surface.
[0005] Furthermore, the stitching typically provides discharge only to the 30 mega ohms level of resistance. To lower this level of resistance, increased stitching through the insole is required. This in turn exacerbates the issue of discomfort. Other designs have embedded conductive material in a foam cushion layer of an insole, but this necessitates holes in the fabric cover layer on top to allow the passage of static electricity to the foam. These holes must be large to create the required lower resistance, further compromising the comfort of the insole.
[0006] An additional issue with existing designs is that they do not typically provide a heel cup and arch support or other smooth, shaped contour to the bottom of the wearer's foot. Moreover, the volume within work footwear is sometimes reduced due to the impact protection integrated into such footwear (e.g., protective toe box). Currently available cushioned electrostatic discharge insoles can take up too much volume, particularly in the toe region of the footwear.
[0007] Prior art US-20230413942-A1 proposes an electrically dissipative removable insole with a lower layer made of an electrically dissipative material and an upper layer made of natural or synthetic fibers. The upper layer has openings that expose the material and allow direct contact with the sole of the wearer's foot. However, this prior art introduces discontinuities in the top cover of the insole. It also does not address the need for a heel cup and arch support. Nor does it address the issue of volume reduction in the toe region.
[0008] Prior Art US-8914996-B2 proposes an antistatic insole with a cover layer, two layers of body, one arch pad, and one heel pad. The first layer is made of rubber and conductive ethylene-vinyl acetate (EV copolymer) to form a trapezoidal shape from the forefoot area to the heel area. While this design includes an arch pad and heel pad but it overcomplicates the assembly into multiple parts that are not integrated together. Thus, the static dissipation is reduced and the cost is increased. It also does not address the volume issues that are sometimes inherent in work footwear.
[0009] EP-3744206-B1 proposes a multilayer (antistatic and/or conductive) insole composed of two layers of fabric, one of which is non-conductive and the other is positioned below it. An

electrically conductive resin is applied to the first layer and passes through the first layer to allow the passage of electric charges. However, the resin reduces the breathability of the top cover layer. The design also does not have the contour to follow the shape of the wearer's foot. It also may take up too much volume in particular footwear.

[0010] Therefore, there is a need to overcome the problems discussed above. The ideal insole would provide an efficient electrostatic discharge pathway without the need for conductive threads stitched through the insole or large holes in the fabric cover layer. It would also provide a heel cup and arch support, and it would not take up too much volume in the toe region of the footwear.

SUMMARY OF THE INVENTION

[0011] The primary objective of the present invention is to provide an insole for electrostatic discharge footwear that offers a comfortable orthotic with low resistance to electrostatic discharge.

[0012] Another objective of the present invention is to incorporate conductive threads woven integrally into the fabric cover of the insole, providing substantial electrical flow paths from the foot of the wearer to the foam layer.

[0013] Yet another objective of the present invention is to construct the foam layer of the insole with an electrostatic discharge (ESD) chemical additive, creating polarity channels for an electrical pathway within the foam layer.

[0014] Still another objective of the present invention is to design the insole in a 3D supportive shape for the foot, enhancing comfort and increasing foot interface, thereby improving the ESD capability of the insole.

[0015] A further objective of the present invention is to mold the bottom of the base of the insole in a way that it better accommodates various footwear, particularly work footwear, without taking up much volume.

[0016] According to one aspect of the present invention, the insole includes a base foam layer and a fabric top cover with conductive threads woven into the cover material. This configuration overlays the base foam layer and provides electrical flow paths from a foot of a wearer to the base foam layer. The fabric top cover can be constructed of a breathable knit material, a woven material, or a non-woven material. The base foam layer can be constructed of various materials, with polyurethane closed-cell foam being favored.

[0017] According to another aspect of the present invention, an electrostatic discharge chemical additive is added to the base foam layer. This additive can be added to a liquid polyurethane material before it is molded, foamed, and cured. Alternatively, conductive particles such as carbon powder may be used. The quantity of the additive can be adjusted to reach desired levels of low-resistance discharge.

[0018] In another embodiment of the present invention, the insole is formed into a 3D supportive shape for a foot for increased comfort and foot interface. This 3D supportive shape may include a heel cup and an arch support. The edges adjacent the toe region of the insole are beveled to accommodate various footwear. The thickness of the region of the insole beneath the toes of the foot can also be reduced to accommodate various footwear.

[0019] According to another aspect of the present invention, the bottom surface of the base foam layer includes molded-in structures that provide relatively more or less cushioning to the foot in various regions of the foot. These molded-in structures can comprise ridges and recesses, posts and ribs.

[0020] The present invention also provides a method for creating an insole for an electrostatic discharge footwear. The method includes providing a base foam layer, providing a fabric top cover having conductive threads woven into the cover material, overlaying the fabric top cover on the base foam layer to provide electrical flow paths from a foot of a wearer to the base foam layer, and adding an electrostatic discharge chemical additive to the base foam layer to create polarity channels for an electrical pathway within the foam layer.

In a further embodiment, a support member is provided to nest below the foam cushion layer. The

support member is a preferably a rigid or semi-rigid member that surrounds at least a portion of the bottom of heel cup. The support member may also underly at least a portion of the arch support. The invention also includes footwear that incorporates the ESD insole and integrates therewith.
[0021] The invention offers several advantages including enhanced comfort, improved interface with the foot, and efficient electrostatic discharge. The insole can be formed into a 3D supportive shape for the foot, increasing comfort and foot interface. The beveled edge of the toe region of the cushion layer of the base better accommodates work footwear. The foregoing paragraphs have been provided by way of general introduction, and are not intended to limit the scope of the following claims. The described embodiments, together with further advantages, will be best understood by reference to the following detailed description taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] Preferred and alternative examples of the present invention are described in detail below with reference to the following drawings:
[0023] FIG. **1** illustrates a footwear insole with electrostatic discharge properties including a cushion member, and a fabric top cover.
[0024] FIG. **2** shows a cross-sectional view of the insole positioned within a shoe, emphasizing the interaction between the insole base and the shoe's midsole and/or outsole.
[0025] FIG. **3** depicts a cross-sectional side-elevational view of the insole, showcasing a top cover and a cushion base with formed recesses and ribs on the bottom for differential cushioning.
[0026] FIG. **4**A presents a top view of the insole with an ESD fabric top cover and contoured heel and arch to fit the wearer's foot shape.
[0027] FIG. **4**B illustrates the fabric top cover material, both the top face and the bottom face.
[0028] FIG. **5** displays the bottom view of an insole featuring a heel cup, arch support, beveled toe edges, and a base cushion structure with ribs and recesses.
[0029] FIG. **6**A shows the bottom view of an insole with a beveled toe edge and various structural features.
[0030] FIG. **6**B shows a toe-end portion of the bottom view of the insole.
[0031] FIG. **6**C shows the toe-end portion in cross section.
[0032] FIG. **7** illustrates the interior of a shoe showing the last board with conductive stitching to create a conductive path from the insole to the midsole or outsole.
[0033] FIG. **8** illustrates an alternate embodiment having a further support structure nested with the foam cushion.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
[0034] Aspects of the present invention are best understood by reference to the description set forth herein. It should be understood, however, that the following descriptions, while indicating preferred aspects and numerous specific details thereof, are given by way of illustration only and should not be treated as limitations. Changes and modifications may be made within the scope herein without departing from the spirit and scope thereof, and the present invention herein includes all such modifications.
[0035] The present invention relates to an insole for an electrostatic discharge (i.e., antistatic) footwear. The insole comprises a base foam layer. This layer can be made from a variety of materials, including but not limited to polyurethane, rubber, or silicone. In the preferred embodiment, the foam layer serves as the primary structure of the insole, providing support and comfort to the wearer's foot.
[0036] The insole further comprises a fabric top cover. This cover preferably includes conductive

threads woven into (or otherwise integrated into) the cover material. These threads create an electrical flow path from the foot of the wearer to the base foam layer. This feature is particularly beneficial in environments where electrostatic discharge is a concern, such as in certain industrial or laboratory settings.

[0037] The fabric top cover can be constructed of a breathable knit material. This material allows for air circulation, helping to keep the wearer's foot dry and comfortable. The conductive threads can be woven into the fabric in a variety of patterns and densities, depending on the desired level of conductivity. Alternatively, the fabric top cover can be a woven material. This type of material can provide a different feel and level of comfort to the wearer. The conductive threads can be woven or otherwise integrated into the fabric in a variety of ways, depending on the desired properties of the insole. In another embodiment, the fabric top cover can be a non-woven material. This type of material can offer a different texture and level of durability compared to woven materials. The conductive threads can be integrated into the non-woven material in a variety of ways, depending on the desired properties of the insole.

The electrical flow paths in the fabric top cover extend from side to side and front to back of the insole. This configuration ensures that the entire surface of the foot is in contact with the conductive threads, providing a consistent and reliable electrical flow path to the foam layer.

[0038] The base foam layer of the insole can be constructed of polyurethane closed- cell foam. This type of foam is known for its durability and resilience, making it an excellent choice for an insole material. The foam can be molded into a variety of shapes and sizes to fit a wide range of footwear. The density of the foam can be adjusted to provide good support to the underside of the foot of the wearer while still providing cupping of the heel and cushioning under the entire foot.

[0039] The electrostatic discharge chemical additive is added to the base foam layer. This additive is mixed into the liquid polyurethane material before it is molded, foamed, and cured. The additive creates polarity channels within the foam, providing an electrical pathway within the foam layer.

[0040] In some embodiments, the additive can be conductive particles such as carbon powder. These particles can be mixed into the liquid polyurethane material before it is molded, foamed, and cured. The particles polarize to discharge static electricity through the foam layer, further enhancing the insole's electrostatic discharge properties.

[0041] The insole can be formed into a 3D supportive shape for the foot. This shape can include features such as a heel cup and an arch support, providing increased comfort and foot-to-insole interface. The 3D shape can be customized to fit the wearer's foot, providing a personalized fit and feel.

[0042] The edges adjacent the toe region (toe and ball of the foot area generally) of the insole can be beveled. This feature allows the insole to better accommodate various types of footwear, including work boots and other types of protective footwear. The beveled edges reduce the overall volume of the insole, allowing it to fit more comfortably within the shoe.

[0043] In some embodiments all or large portions of the thickness of the region of the insole beneath the toes of the foot can be reduced. Again, this feature allows the insole to better fit within shoes with a smaller toe box, such as work boots or other types of protective footwear.

[0044] The bottom surface of the base foam layer can include molded-in structures. These structures can provide more or less cushioning to the foot in various regions, depending on the wearer's needs. This feature allows for a more customized level of comfort and support even with a dense foam cushion that provide heel cupping and arch support.

[0045] The molded-in structures can include ridges and recesses. These features can be strategically placed to provide additional support or cushioning in specific areas of the foot. For example, ridges can be placed in areas that need more support, such as the arch, while recesses can be placed in areas that need more cushioning, such as the heel. An area needing more support would have more space between ridges, for example.

[0046] The present invention also provides a method for creating an insole for an electrostatic

discharge footwear. This method involves providing a base foam layer. The foam layer can be created using a variety of materials and methods, depending on the desired properties of the insole.
[0047] The method further involves providing a fabric top cover with conductive threads woven into the cover material. The conductive threads can be woven into the fabric using a variety of techniques, depending on the desired properties of the insole.
[0048] The fabric top cover is then overlaid on the base foam layer. This step creates an electrical flow path from the foot of the wearer to the base foam layer. The overlaying process can be done using a variety of methods, depending on the desired properties of the insole.
[0049] The base foam layer can be constructed of polyurethane closed-cell foam. This type of foam can provide a high level of support and comfort to the wearer. The foam can be molded into a variety of shapes and sizes to fit a wide range of footwear. The density of the foam can be selected to retain supportive shape, especially to the region of the insole rearward of the toe region, such as the arch and heel region.
[0050] The electrostatic discharge chemical additive is added to the base foam layer. This additive is mixed into the liquid polyurethane material before it is molded, foamed, and cured. The additive creates polarity channels within the foam, providing an electrical pathway within the foam layer.
[0051] In some embodiments, the additive can be conductive particles such as carbon powder. These particles can be mixed into the liquid polyurethane material before it is molded, foamed, and cured. The particles polarize to discharge static electricity through the foam layer, further enhancing the insole's electrostatic discharge properties.
[0052] The method can further involve forming the insole into a 3D supportive shape for a foot. This shape can include features such as a heel cup and an arch support, providing increased comfort and foot interface. The 3D shape can be customized to fit the wearer's foot, providing a personalized fit and feel.
[0053] The edges adjacent the toe region of the insole can be beveled as part of the forming process. This feature allows the insole to better accommodate various types of footwear, including work boots and other types of protective footwear. The beveled edges reduce the overall volume of the insole, allowing it to fit more comfortably within the shoe.
[0054] The thickness of the region of the insole beneath the toes of the foot can be reduced as part of the forming process. This feature allows the insole to better fit within shoes with a smaller toe box, such as work boots or other types of protective footwear. The reduced thickness does not compromise the comfort or support provided by the insole. Cushion thickness under the ball of the foot can be maintained, while it can taper forward of the ball of the foot towards the front of the insole where a beveled edge can further reduce thickness.
[0055] The bottom surface of the base foam layer can include molded-in structures that provide relatively more or less cushioning to the foot in various regions of the foot. These structures can be created using a variety of methods, depending on the desired properties of the insole.
[0056] The present invention provides a supportive, comfortable insole that can meet or be well below required maximum levels of resistance to electrostatic discharge. The insole interfaces well with a variety of footwear, making it a versatile solution for a wide range of applications. The insole's unique combination of features, including its conductive threads, foam layer, and molded-in structures, provide a high level of comfort and support while also effectively discharging static electricity.
[0057] Referring now to FIG. **1**, it illustrates a footwear insole **100** with electrostatic discharge properties. The insole **100** includes a base **102** and a top cover **106**. The base includes a cushion **104**. The base may also include support semi-rigid or rigid structures to support the cushion. The insole **100** is designed to provide both comfort and low resistance to electrostatic discharge. The top cover **106** overlays the cushion **104** and provides substantial electrical flow paths from the foot of the wearer to the bottom of the base **102**. The base **102** is designed to be comfortable and supportive, with a heel cup **108** and an arch support **110** shaped to the bottom of the wearer's foot.

The toe region of the insole **100** includes a beveled edge **114** to better accommodate various types of footwear.

[0058] The cushion layer **104** of the insole **100** as depicted in FIG. **1**, can be constructed of various materials, with a preferred material being polyurethane closed-cell foam. This foam is favored due to its comfort and supportive properties. The foam is elastomeric. The durometer of the elastomeric foam is selected to provide the right balance of support and cushion. Preferably, the durometer is approximately **44***c*, based on the Asker C scale. With the preferred construction, relying on the foam for heel cup and arch support, the range of durometer is between about **41**-**47***c*. Higher or lower durometer targets are used on other embodiments depending on the construction, shape, contour, and overall thickness of the insole. Refer further to the discussion below with regard to FIG. **8** where an additional support member might allow the foam to have a lower durometer while still providing good support. Likewise, the density or specific gravity of the polyurethan closed-cell foam is preferably in the range of 0.33+/−0.02. Higher or lower densities may be used, again depending on the structure, shape, contour, and overall thickness of the insole product.

[0059] The foam layer of the base **102** is designed to be molded, foamed, and cured with an electrostatic discharge (ESD) chemical additive. This additive creates polarity channels within the foam, providing an electrical pathway for the discharge of static electricity.

[0060] The top cover **106** of the insole **100** as shown in FIG. **1**, is constructed with conductive threads woven integrally into the cover material. The threads run throughout the cover material from side to side and front to back, providing ample electrical flow paths from the wearer's foot to the base **102**. The top cover **106** is preferably constructed of a breathable knit material, but it may also be a woven or other non-woven material.

[0061] As seen in FIG. **1**, the heel cup **108** and the arch support **110** are part of the 3D supportive shape of the insole **100**. These features are designed to provide increased comfort and foot interface. The heel cup **108** cradles the heel of the wearer, while the arch support **110** provides support to the arch of the foot. This increased foot interface enhances the ESD capability of the insole **100**. Thus, the top cover **106** may be made with a lower amount of conductive material to achieve the antistatic electrical conduction therethrough.

[0062] The beveled edge **114** in the toe region of the insole **100** as illustrated in FIGS. **1**, **5**, and **6**, is designed to better accommodate various types of footwear, particularly work footwear. The protective features of work footwear often create a decreased volume within the toe box. The beveled edge **114** allows the insole **100** to better integrate within the footwear without taking up much volume. The thickness of the region of the insole **100** beneath the toes of the foot (toe region **112**) can also be reduced to accommodate such footwear as further detailed below in connection with FIGS. **6**A-C.

[0063] Referring to FIG. **2**, a cross-sectional view of the insole **100** positioned within a shoe (footwear **200**) is depicted, emphasizing the interaction between the insole base **102** and the shoe's midsole **204** and/or outsole **202**. The figure provides a detailed view of footwear **200**, the insole **100**, the midsole **204**, and the outsole **202**. The footwear **200** includes the insole **100**, the midsole **204**, and the outsole **202**. The insole **100** is positioned within the shoe and rests on the midsole **204** or a last board (shown in FIG. **7**).

[0064] The base **102** of the insole is designed to be comfortable and supportive. It is shaped to the bottom of the wearer's foot, providing a heel cup **108** and an arch support **110**. This contouring not only enhances comfort but also decreases the insole's resistance to electrostatic discharge due to increased engagement with the wearer's foot. The base **102** of the insole is preferably constructed with a cushion **104** made from a foam material, preferably polyurethane closed-cell foam, which is molded, foamed, and cured with an electrostatic discharge (ESD) chemical additive. This additive creates polarity channels within the foam, providing an electrical pathway. The base **102** may also have a semi-rigid support structure beneath and/or beside at least a portion of the cushion **104** to increase support to cushion **104** and insole **100**.

[0065] The midsole **204** is a component of the shoe that lies between the insole **100** and the outsole **202**. Often the footwear also includes a last board on top of or in place of the midsole. The midsole **204** provides additional support and cushioning to the wearer's foot. It is typically softer than the outsole **202** while the outsole may be more durable. The insole **100**, specifically the base **102** of the insole, interacts with the midsole **204**, transmitting the electrical flow from the insole **100** to the midsole **204**. This interaction is facilitated by the conductive properties of the insole **100**, particularly the conductive threads in the fabric top cover and the ESD chemical additive in the base foam layer. As further explained below, the footwear **200** might also include a Strobel or last board between the insole and the midsole or outsole.

[0066] The outsole **202** is the bottom part of the shoe that comes in direct contact with the ground. The outsole **202** is designed to be durable and provide traction. The insole **100**, particularly the base **102** of the insole, also interacts with the outsole **202**, transmitting the electrical flow from the insole **100** to the outsole **202** through the last board and midsole, if included in the footwear. This interaction further enhances the shoe's low resistance to electrostatic discharge, making the footwear **200** safe for static-sensitive work zones.

[0067] FIG. **3** illustrates a cross-sectional side-elevational view of the insole **100**. The base **102** and top cover **106** are shown. The base **102** includes cushion **104**. The upper side of cushion **104** is generally contoured to the shape of the foot. The bottom side of the cushion **104** (or base **102**) and has formed recesses **302** and ribs **300** for differential cushioning. The top cover **106** overlays the base **102**, providing a smooth surface for the foot of the wearer.

[0068] FIG. **4**A presents a top view of the insole with an ESD fabric top cover **106** and contoured heel cup **108** and arch support **110** designed to fit the wearer's foot shape. The ESD fabric top cover **106** overlays a base foam layer (not shown in this figure) and is woven with conductive threads that provide electrical flow paths from the foot of the wearer to the foam layer. The contoured heel cup **108** and arch support **110** are designed to provide both comfort and lower ESD resistance due to more engagement with the wearer's foot.

[0069] FIG. **4**B shows the fabric top cover **106** prior to being molded with the cushion **104**. Once molded with the cushion **104**, the top cover **106** is trimmed, preferably by die cutting the fabric to match the shape of the cushion **104** and as shown in FIG. **4**A. The top cover **106** includes a top face **402** and a bottom face **404**. Bottom face **404** includes a film **406**. Film **406** helps bond cushion **104** to top cover **106** while limiting penetration of the uncured cushion material (polyurethan liquid base) through the fabric of top cover **106**. In a preferred embodiment, film **406** is a thin polyurethane film layer applied only to the bottom of top cover **106**. It functions as a bonding layer for the cushion polyurethane to adhere to during the foaming and curing process. No additional adhesives are required during the construction of the insole. Such adhesives could compromise the electrostatic discharge properties. The film **406** used has a very low electrical resistance so as to not significantly impact the overall static discharge of the insole. The thickness is preferably about 0.02 mm and can be in the range of 0.10 mm to 0.01 mm.

[0070] Referring now to FIGS. **5** and **6**A-C, the bottom of the inventive insole **100** is illustrated. The figures show the various structural features of the insole, including the insole bottom, a beveled toe edge **114**, and a tread pattern **500**. These features are designed to provide comfort to the wearer and accommodate different types of footwear. As previously noted, the insole **100** features a heel cup **108**, arch support **110**, beveled toe edges **114**. The insole also includes a base cushion structure or tread pattern **500** with ribs **300** and recesses **302**. The insole **100** is designed to provide a comfortable orthotic with low resistance to electrostatic discharge.

[0071] The base **102** provides the overall shape and support for the insole **100**. As described previously, it is constructed from a foam layer. The ribs **300** and recesses **302** are molded into the bottom surface of the base cushion structure **500**. These structures provide varying levels of cushioning to different regions of the foot, allowing for a customized level of comfort and support. For example, variations for heel strike and toe push may be made. The base structure **500** can also provide extra friction between the bottom of the insole **100** and the last board or midsole. This will help hold the insole in place for better foot stability.

[0072] The tread pattern **500** is molded into the bottom surface of the base cushion layer. The tread pattern can include various ridges and recesses that are strategically placed to provide more or less cushioning to different regions of the foot, such as the heel strike and toe push areas. This feature enhances the insole's comfort and support, while also improving its interface with the foot.

[0073] The beveled toe edge **114**, shown in these figures, enhances its compatibility with various types of footwear. The beveled edge allows the insole to better fit within the toe box **208** of the shoe **200**, especially in work footwear where protective features often decrease the available volume. This design consideration ensures that the insole can provide support and comfort without taking up excessive space.

[0074] FIG. **6**B shows a sulcus line **602** between the regions of the insole that correspond to the ball of the foot and the toes of the foot. Preferably, the thickness of the insole rearward (towards the heel) of the sulcus line **602** is approximately 5.0 mm. A slight taper of the bottom surface of the cushion **104** then extends to the toe end of the insole such that the thickness of the insole is approximately 3.1 mm. Thus, the reduction in thickness extends forward from the sulcus line **602** to the toe end of the insole at the beveled edge **114** and a rounded tip **608**. the progression of the taper is shown by lines **610**. This taper increases the volume within the toe box of the footwear-a place where volume might be reduced from protective toe-cap features of work footwear. However, the cushion thickness in the region of the ball of the foot (to just past where the metatarsal heads press onto the insole) remains significant, promoting foot comfort. The beveled edge **114** is a reduction in addition to the thickness taper.

[0075] FIG. **7** illustrates the interior of a shoe, denoted by reference numeral **200**. The shoe **200** includes a last board **700**, which is shown with conductive stitching **702**. The conductive stitching **702** creates a conductive path from the insole to the midsole **204** or outsole **202**. The last board **700** is an integral part of the shoe **200** and serves as the foundation on which the shoe is constructed. It is typically made from a durable material that can withstand the wear and tear of daily use.

[0076] The conductive stitching **702** is woven into the last board **700**. The stitching **702** is composed of conductive threads that are designed to provide a path for electrical flow. Since the insole **100** with cushion **104** is layered above the last board **700**, the threads sewn through the last board are not felt by the wearer. The density of the electrical threads (conductive stitching **702**) through the last board can be provided to achieve the desired conductive flow from the insole **100** to the midsole and or outsole. Other means of providing a conductive last board are also possible.

[0077] The midsole **204** and the outsole **202** are the parts of the shoe **200** that come into direct contact with the ground. The midsole **204** is typically made from a material that can absorb shock and provide cushioning, while the outsole **202** is designed to provide traction and resist wear. The conductive path created by the conductive stitching **702** extends from the insole, through the last board **700**, and to the midsole **204** and/or the outsole **202**. This allows for the discharge of static electricity, thereby enhancing the ESD capability of the shoe **200**.

FIG. **8** shows a further embodiment that includes a support member **800**. Support member **800** is more rigid than cushion **104**. Support member **800** may be, for example, a plastic material rather than a foam material. Support member **800** cradles cushion **104** to hold it in place and still provide support even with a less dense foam in cushion **104**. However, as shown in FIG. **8**, at least half of the bottom surface of base **102** exposes cushion **104**. Thus, support member **800** does not need to be discharge conductive to provide adequate dissipation of static electricity. Support member **800** preferably surrounds heel cup **108** and underlies at least a portion of arch support **110** of cushion **104**.

[0078] The U-shape of support member **800** exposes a central region of cushion **104** beneath heel cup **108**. This provides conducive dissipation over a large area and also allows support member **800** to flex inwardly or outwardly to accommodate footwear of various widths. Of course, support

member **800** may also include conductive material therein to provide static electricity dissipation.
[0079] The insole designed as per the innovative aspects of the present invention is not just functionally superior, but also excels in providing user comfort. In some embodiments, the insole is formed into a 3D supportive shape for the foot. This shape includes design elements such as a heel cup and an arch support. These features are intended to enhance the comfort of the wearer and to increase foot interface, thereby naturally improving the ESD capability of the insole.
[0080] The present invention also addresses the common issue of footwear compatibility. In certain embodiments of the insole, the edges adjacent to the toe region are beveled. This design element allows the insole to better fit within the toe box of various types of footwear, especially work footwear where the protective features often decrease the available volume. Similarly, the thickness of the region of the insole beneath the ball and toes of the foot can be reduced to accommodate such footwear, ensuring the comfort of the wearer is not compromised.
[0081] The bottom surface of the base cushion layer of the insole may also be tailored to enhance comfort. In certain embodiments, the bottom surface includes molded-in structures that provide relatively more or less cushioning to the foot in the various regions. For example, the structures may take the form of ridges and recesses, strategically placed to create regions that provide more or less cushioning. This allows for a customized level of comfort and support to the wearer, further enhancing the usability of the insole.
[0082] Compared to the prior art solutions, the present invention offers several advantages including its enhanced comfort, improved interface with the foot, and efficient electrostatic discharge. The contoured 3D shape for the foot, the conductive threads in the fabric top layer, and the ESD chemical additive in the base foam layer work synergistically to create an insole that is both comfortable and effective in reducing ESD resistance. Additionally, the beveled edge of the toe region of the cushion layer of the base and the adjustable thickness beneath the ball and toes of the foot better accommodates work footwear, making this insole a versatile solution for all types of footwear.
[0083] The embodiments of the present invention disclosed herein are intended to be illustrative and not limiting. Other embodiments are possible, and modifications may be made to the embodiments without departing from the spirit and scope of the invention. As such, these embodiments are only illustrative of the inventive concepts contained herein. It should be understood that the present invention can be practiced with modification and alteration within the spirit and scope of the appended claims. The invention should not be limited to the described and depicted embodiments, but should be defined by the appended claims, which therefore include all equivalents of the disclosed and claimed invention.

Claims

1. An insole for electrostatic discharge footwear comprising: a base foam layer having a unitary body with a contoured upward-facing shape to follow the shape of a wearer's foot; a fabric top cover having conductive threads woven into the cover material, overlaying the base foam layer and providing electrical flow paths from a foot of a wearer to the base foam layer; and an electrostatic discharge chemical additive added to the base foam layer, creating polarity channels for an electrical pathway within the foam layer.
2. The insole of claim 1, wherein the fabric top cover is constructed of a breathable knit material.
3. The insole of claim 1, wherein the fabric top cover is a woven material.
4. The insole of claim 1, wherein the fabric top cover is a non-woven material.
5. The insole of claim 1, wherein the base foam layer is constructed of polyurethane closed-cell foam, the foam having a durometer in the range of **41***c* to **47***c* on the Asker c scale.
6. The insole of claim 1, wherein the electrostatic discharge chemical additive is added to a liquid polyurethane material before it is molded, foamed, and cured.

7. The insole of claim 1, further comprising a semi-rigid support member along at least a portion of a bottom of the base foam layer, the semi-rigid support member providing additional heel and arch support to the base foam layer.
8. The insole of claim 7, wherein the support member includes a conductive material.
9. The insole of claim 7, wherein the support member covers less than half of a lower surface area of the base foam layer.
10. An insole for an electrostatic discharge footwear, the insole being formed in a 3D supportive shape for a foot for increased comfort and foot interface, comprising: a base foam layer; and a fabric top cover having conductive threads woven into the cover material, overlaying the base foam layer and providing electrical flow paths from a foot of a wearer to the base foam layer.
11. The insole of claim 10, wherein the **3**D supportive shape comprises a heel cup and an arch support.
12. The insole of claim 10, wherein the edges adjacent the toe region of the insole are beveled to accommodate various footwear.
13. The insole of claim 10, wherein the thickness of the region of the insole beneath the ball and toes of the foot is reduced to accommodate various footwear.
14. The insole of claim 10, wherein the bottom surface of the base foam layer includes molded-in structures that provide relatively more or less cushioning to the foot in various regions of the foot.
15. The insole of claim 14, wherein the molded-in structures comprise ridges and recesses.
16. A method for creating an insole for an electrostatic discharge footwear, the method comprising: providing a base foam layer with a conductive additive; providing a fabric top cover having conductive threads woven into the cover material; and overlaying the fabric top cover on the base foam layer to provide electrical flow paths from a foot of a wearer to the base foam layer.
17. The method of claim 16, wherein the fabric top cover is constructed of a breathable knit material.
18. The method of claim 16, wherein the fabric top cover is a woven material.
19. The method of claim 16, wherein the base foam layer is constructed of polyurethane closed-cell foam having a specific gravity between 0.31 to 0.35 to provide contoured foot support.
20. The method of claim 16, wherein the electrostatic discharge chemical additive is added to a liquid polyurethane material before it is molded, foamed, and cured.
21. The method of claim 16, wherein the additive is conductive particles such as carbon powder.
22. The method of claim 16, further comprising forming the insole into a 3D supportive shape for a foot for increased comfort and foot interface.
23. The method of claim 22, wherein the 3D supportive shape comprises a heel cup and an arch support.
24. The method of claim 22, wherein the edges adjacent the toe region of the insole are beveled to accommodate various footwear.
25. The method of claim 22, wherein the thickness of the region of the insole beneath the ball and toes of the foot is reduced to accommodate various footwear.
26. The method of claim 22, wherein the bottom surface of the base foam layer includes molded-in structures that provide relatively more or less cushioning to the foot in various regions of the foot.
27. Footwear having electrostatic discharge properties, the footwear comprising: an upper to surround at least a portion of the foot of a wearer; an outsole engaged to the upper, the outsole having electrostatic discharge properties embedded therein; an insole comprising: a cushion member having conductive material therein, the conductive material allowing transmission of static electricity from the top of the cushion member to the bottom of the cushion member; a cover material fixed to the cushion member, the cover material being a fabric, the fabric having fabric threads; wherein the fabric includes conductive threads extending through the fabric from top to bottom, the conductive threads being integrated with the fabric threads.
28. The footwear of claim 27, further comprising: a midsole secured to a top of the outsole, the

midsole having electrostatic discharge properties embedded therein; and a last board secured to a top of the midsole, the last board having electrostatic discharge properties for discharging static electricity from the insole to the midsole; wherein the insole rests on the midsole.

29. The footwear of claim 28, wherein the cushion member comprises a closed-cell foam having an additive providing electrostatic discharge properties, the cushion member having an upper surface contour substantially following the bottom contour of a foot.

30. The footwear of claim 29, further comprising a semi-rigid support member along at least a portion of a bottom of the cushion member, the semi-rigid support member providing additional heel and arch support to the cushion member.

31. The footwear of claim 28, wherein the last board includes a thread extending therethrough from top to bottom, the thread being conductive.